

SEREN

Every child is a star. Every adult deserves a second chance.

Complete Master Research & Technical Dataset Blueprint Problem Landscape · Full 80+ Condition Catalogue · Detection Science · Verified Open Dataset Catalogue · Condition-to-Dataset Mapping · Training Protocols

The complete, single-document scientific and technical foundation for SEREN's AI-powered, offline-first mobile screening and intervention platform — covering 80+ neurodevelopmental, learning, communication, emotional, sensory, and confidence-based conditions across childhood, adolescence, and adulthood.

84

CONDITIONS
MAPPED TO
DATA

45+

VERIFIED OPEN
DATASETS

8

AI MODEL
MODULES

250M

INDIAN
CHILDREN IN
SCOPE

Founder: Sansakardeep Talikote

CONFIDENTIAL — INTERNAL RESEARCH DOCUMENT — COMPLETE SINGLE-VOLUME EDITION

Table of Contents

1. Executive Overview	
2. The Problem Landscape — Scale & Policy Context	
3. Complete Condition Catalogue — 80+ Conditions, 8 Categories	
4. Flagship Condition Profiles — Neuroscience & Detection Signals	
5. SEREN AI Architecture — 8-Module Ensemble	
6. Complete Verified Open-Source Dataset Catalogue	
7. Full Condition → Dataset → Detection-Modality Map (84 rows)	
8. Training Protocols — Environment, Download Commands & Model Code	
9. Accuracy Model, Validation Roadmap & Honest Limitations	
10. Reference Bibliography	

1. Executive Overview

SEREN is an AI-powered, offline-first mobile platform that screens for 80+ neurodevelopmental, learning, attention, speech, anxiety, emotional, sensory, and confidence-based conditions across the full human lifespan — ages 5 through 60+ — using only a standard smartphone, and follows every flagged result with a short daily intervention built for that specific condition.

This single document consolidates SEREN's full scientific and technical foundation: the size of the problem, the complete condition catalogue, the neuroscience behind flagship conditions, the AI model architecture, and — the core of this edition — a fully verified catalogue of real, publicly available open-source datasets that can be used to pre-train each of SEREN's detection modules, mapped condition-by-condition, with direct links and exact training use. Every dataset listed here was checked against its original source before inclusion. Nothing in this document is a clinical diagnostic claim: SEREN is a screening instrument, and every positive result is designed to lead to a recommended professional referral, not replace one.

2. The Problem Landscape — Scale & Policy Context

2.1 Scale in India

India's school-age population (ages 5–14) is approximately 250 million — the largest school-age cohort of any country. A systematic review and meta-analysis in the *Indian Journal of Psychiatric Nursing* (Joseph & Devu, 2022) pooled Learning Disability prevalence across Indian studies at 10.70% (95% CI: 7.10–14.3%). A South Indian regional study (*Archives of Mental Health*, 2023) reported dyslexia at 11.2%, dysgraphia at 12.5%, and dyscalculia at 10.5%. Extrapolated conservatively, this places Indian children carrying at least one undetected learning-related condition at well over three crore (30 million+).

2.2 The Adult Burden

Dyslexia, ADHD, anxiety, and stuttering persist well past age 18. Research on adults with unidentified ADHD documents measurably lower average income relative to peers (Fayyad et al., 2007). A 2024 study (PMC) showed that a structured web-based linguistic intervention over 24 sessions of 15–20 minutes produced measurable literacy improvement in adults aged 16–30 with dyslexia — direct evidence for SEREN's adult module.

2.3 Policy Context — NEP 2020 & NIPUN Bharat

India's National Education Policy (NEP) 2020 designates Foundational Literacy and Numeracy (FLN) as the school system's highest priority, operationalised through the NIPUN Bharat mission targeting universal FLN by Grade 3, with explicit provisions for identifying children with learning difficulties in mainstream classrooms. Current implementation relies mostly on population-level literacy surveys; SEREN is designed as the individual-level, condition-specific screening layer that complements these tools.

3. Complete Condition Catalogue — 80+ Conditions, 8 Categories

Category	#	Conditions
1. Learning Disorders	9	Dyslexia • Dyscalculia • Dysgraphia • Dyspraxia/DCD • Irlen Syndrome/Visual Stress • Non-Verbal Learning Disability (NVLD) • Reading Fluency Lag • Orthographic Processing Deficit • Undiagnosed Adult Dyslexia
2. Attention & Processing	9	ADHD Inattentive • ADHD Hyperactive/Impulsive • ADHD Combined • Auditory Processing Disorder • Processing Speed Disorder • Working Memory Deficit • Time-Blindness • Executive Function Deficit • Undiagnosed Adult ADHD
3. Speech & Language	10	Stuttering • Word-Finding Difficulty • Expressive Language Disorder • Receptive Language Disorder • Social Communication Disorder • Childhood Apraxia of Speech • Phonological Disorder • Cluttering • Voice Disorder • Accent-Related Communication Anxiety
4. Anxiety Disorders	11	Generalised Anxiety Disorder • Social Anxiety Disorder • Selective Mutism • Exam/Test Anxiety • Separation Anxiety • School Refusal • Specific Phobia • Panic Disorder • Performance Anxiety • Illness-Related Health Anxiety (Child) • Health Anxiety (Adult)
5. Emotional & Behavioural	10	Childhood Depression (Masked) • Emotional Dysregulation • Oppositional Anxiety-Driven Behaviour • Adolescent Depression • Adjustment Disorder • Post-Disruption Learning-Loss Anxiety • Burnout Syndrome (Adult) • Parental Burnout • Trauma-Based Silence • Grief-Related Withdrawal
6. Silent Profiles & Neurodiversity	11	Masking/Camouflaging • High-Functioning Autism (Masked) • Social Communication Disorder (isolated) • Twice-Exceptional Profile (2e) • Pathological Demand Avoidance • Gifted Underachievement • Introversion-Driven Suppression • Invisible Struggle • Alexithymia • Highly Sensitive Person Overwhelm • Chronic People-Pleasing
7. Sensory & Motor	4	Sensory Processing Disorder • Sensory Defensiveness • Vestibular Processing Difficulty • Proprioceptive Processing Difficulty
8. Insecurities & Confidence Deficits	21	Body Image Insecurity • Imposter Syndrome • Rejection Sensitivity • Workplace/Career Insecurity • Relationship Insecurity • FOMO/Social Comparison Anxiety • Digital/Online Social Insecurity • Academic Trauma • Stuttering-Linked Confidence Deficit • Decision Paralysis • Financial Confidence Insecurity • Leadership Avoidance • Career-Stagnation Collapse • Deep-Rooted Shyness • Voice/Expression Insecurity • Social Belonging Insecurity • Performance/Achievement Insecurity • Anger as Hidden Insecurity • Family-Origin Insecurities • Perfectionism-Driven Self-Criticism • Public Speaking Phobia

4. Flagship Condition Profiles — Neuroscience & Detection Signals

Dyslexia — Learning Disorders (5-20% global, target accuracy 95-99%)

Basis: phonological processing deficit; reduced left-hemisphere reading-network activation (temporoparietal junction, visual word form area, Broca's area); heritability ~50-60%.

Signs: below-grade reading despite adequate instruction; letter reversals past expected age; slow laborious oral reading; weak phoneme blending/rhyme; strong listening vs. weak reading comprehension.

Detection signals: eye-tracking fixation/regression/saccade metrics; RAN latency; phoneme segmentation accuracy; handwriting-CNN reversal rate; oral reading fluency; sight-word latency.

ADHD - Inattentive Type — Attention & Processing (7.2% global, target 90-95%)

Basis: reduced prefrontal/basal ganglia/cerebellar volume; dopaminergic-noradrenergic dysregulation; heritability ~75-80%; under-identified in girls and high-ability children.

Signs: difficulty sustaining attention; incomplete multi-step tasks; poor time estimation; in adults — chronic disorganisation and missed deadlines.

Detection signals: CPT miss rate; vigilance decrement; gaze deviation during instructions; omission-error rate; session-interruption frequency; reaction-time variability.

Stuttering — Speech & Language (~1% global, target 90-95%)

Basis: neurological timing disruption in speech motor planning (basal ganglia-thalamo-cortical circuitry); heritability ~70%; ~80% of childhood cases resolve spontaneously.

Signs: repetitions, prolongations, blocks; secondary behaviours; avoidance vocabulary; anticipatory anxiety before speaking.

Detection signals: disfluency audio classifier; spectral prolongation analysis; intra-word silence thresholding; facial-landmark tracking correlated to speech timing.

Social Anxiety Disorder — Anxiety Disorders (7-13% lifetime, target 87-92%)

Basis: fear of negative social evaluation; right amygdala hyperactivation; peaks in adolescence.

Signs: fear of judgment; avoidance of oral participation; blushing/voice tremor; in adults — avoidance of networking, interviews, phone calls.

Detection signals: gaze-aversion frequency; voice-tremor (jitter/shimmer); response-delay differential solo vs. group; facial micro-expression analysis; camera heart-rate proxy.

Masking and Camouflaging — Silent Profiles (~5% global, target 75-82%)

Basis: active concealment of neurodivergent traits; exhausting; linked to adult burnout; more documented in girls/women.

Signs: typical public performance vs. private struggle; post-performance exhaustion; context-dependent inconsistent performance; preference for silent failure.

Detection signals: accuracy-vs-effort gap; fatigue-curve decline in session's final third; performance variance on repeated identical tasks; unused-hint frequency.

Imposter Syndrome — Insecurities & Confidence (~70% of professionals, target 75-81%)

Basis: persistent internal belief of being less competent than evidenced achievements suggest (Clance & Imes, 1978).

Signs: attributing success to luck; fear of being "found out"; chronic over-preparation; discomfort accepting praise.

Detection signals: NLP sentiment on self-referential success language; help-refusal pattern; success-minimisation phrase detection.

5. SEREN AI Architecture — 8-Module Ensemble

SEREN's AI engine is an ensemble of eight specialised sub-models, each trained on a distinct data modality. Their outputs are fused by a meta-classifier (FusionNet) into final condition risk scores.

Table 5.1 — Eight-module ensemble

Module	Input Modality	Model Type	Primary Conditions
DrawNet	Touchscreen stroke data: velocity, pressure, shape	CNN + LSTM on stroke time-series	Dyslexia (reversals), Dysgraphia, DCD
GazeNet	Eye movement (front camera, MediaPipe iris)	LSTM sequence classifier	Dyslexia (regressions), ADHD (gaze)
PhonNet	Audio + response latency (phonological tasks)	Wav2Vec2 fine-tuned + latency regressor	Dyslexia, APD, Stuttering, Word-Finding
NumNet	Touch reaction-time on number comparison tasks	Random Forest on RT features	Dyscalculia, Processing Speed
AttentNet	Task completion patterns, miss rates, gaze	XGBoost on behavioural features	ADHD, Processing Speed, Anxiety
EmotNet	Teacher questionnaire + task-abandonment signals	Logistic Regression + Rule Engine	Anxiety, Selective Mutism, Withdrawal
SpatialNet	Pattern-completion accuracy + speed	SVM on accuracy-latency features	Dyscalculia, Hidden Strengths, 2e
FusionNet	All 7 sub-model score vectors	Weighted ensemble (Random Forest)	All conditions — final risk output

5.1 Three-Phase Training Pipeline

- **Phase 1 — Global Pre-training:** models initialised on globally available datasets (Section 6). Establishes baseline feature recognition.
- **Phase 2 — Indian Fine-Tuning:** pre-trained models fine-tuned on SEREN pilot-school data (Hindi/Marathi medium). Critical for India-specific accuracy.
- **Phase 3 — Continuous Learning:** anonymised, consent-cleared data aggregated with each new school deployment; accuracy improves monotonically.

All training involving minors follows strict protocol: parental consent obtained; all PII stripped before training; data stored on-device by default; cloud aggregation only with explicit opt-in; DPDP Act 2023 compliant.

6. Complete Verified Open-Source Dataset Catalogue

Every dataset below was checked against its original publisher (Zenodo, Kaggle, Mendeley, GitHub, PhysioNet-class academic archives, or the originating research paper) before inclusion. Where a dataset requires free registration, that is noted under License.

6.1 Dyslexia Datasets

Dataset	Link	License	SEREN Use
ETDD70	zenodo.org/records/13332134	CC BY 4.0	GazeNet pre-training: fixation duration, regression rate (70 children, ages 9–10, eye-tracking)
Dyslexia Handwriting (Kaggle)	kaggle.com/datasets/drizasazanitaisa/dyslexia-handwriting-dataset	Open (Kaggle)	DrawNet: letter-reversal CNN pre-training (138,500 labeled images)
Synthetic Dyslexia HW (YOLO)	zenodo.org/records/14852659	CC BY 4.0	DrawNet: augmented reversal-pattern detection training (130k images)
Dyslexia Prediction (Kaggle)	kaggle.com/datasets/luzrello/dyslexia	CC BY 4.0	FusionNet: cognitive risk-factor weighting (535 participants, gamified test features)
OpenNeuro ds003126	openneuro.org/datasets/ds003126	Open Access	Reference validation only — MRI/fMRI ground truth for reading-network correlation, not on-device inference
CopCo (Danish ET+Dyslexia)	via arxiv.org/pdf/2602.19598	Research	GazeNet: cross-linguistic reading-regression pattern reference (57 participants)

6.2 Dysgraphia Datasets

Dataset	Link	License	SEREN Use
IAM Handwriting Database	fki.inf.unibe.ch/databases/iam-handwriting-database	Free registration	DrawNet: baseline stroke segmentation and character-level features (657 writers)
ICDAR Datasets (2013/2019)	tc11.cvc.uab.es/datasets	Research	DrawNet: large-scale handwriting CNN pre-training, multi-language competitions
CASIA Handwriting Database	nlpr.ia.ac.cn/databases/handwriting	Research	DrawNet: non-Latin stroke-pattern transfer learning
Devanagari Character Dataset	kaggle.com/datasets/suvooo/hindi-character-recognition	CC BY	Critical: Hindi-script letter-formation norms — mandatory pre-training step before Hindi-medium deployment
Dysgraphia AI Review (PLOS ONE 2025)	doi.org/10.1371/journal.pone.0328722	Open Access	Dataset discovery reference — systematic index of dysgraphia AI datasets

6.3 Dyscalculia Datasets

Dataset	Link	License	SEREN Use
PISA Math Dataset	pisa.oecd.org	OECD Open	NumNet: cross-cultural math-difficulty normative benchmark (79 countries)

Number Sense Screener Norms	doi.org/10.1007/s11145-009-9168-7	Research	NumNet: age-stratified normative baseline for dot-counting/subitizing tasks
-----------------------------	---	----------	---

6.4 ADHD & Attention Datasets

Dataset	Link	License	SEREN Use
IEEE EEG ADHD Children	iee-dataport.org/open-access/eeg-data-adhd-control-children	CC BY-NC-ND 4.0	AttentNet: theta/beta-ratio feature reference (61 ADHD + 60 control children, ages 7–12)
Mendeley EEG ADHD Adults	doi.org/10.17632/6k4g25fhzg.1	CC BY 4.0	AttentNet: adult ADHD signal reference for Phase 2 adaptation (79 adults, 4 recording states)
ADHD-200 (fMRI)	fcon_1000.projects.nitrc.org/indi/adhd200	Open Access	Research reference only — SEREN uses behavioural proxies, not fMRI, for on-device inference (973 participants)
Reddit Mental-Health Text	github.com/amurark/mental-health-detection	Open	EmotNet: ADHD/anxiety/depression language-pattern validation from labeled Reddit posts

6.5 Speech, Fluency & Stuttering Datasets

Dataset	Link	License	SEREN Use
UCLASS	uclass.org.uk (free registration)	Free research	PhonNet: stuttering acoustic-feature training (139 monologues, 81 speakers ages 5–47)
FluencyBank	fluency.talkbank.org	CC BY-NC-SA 3.0	PhonNet: fluency-disorder feature extraction, expert-annotated disfluency labels (48 speakers, ~25 hrs)
SEP-28k	github.com/apple/ml-stuttering-events-dataset	CC BY-NC 4.0	PhonNet: largest open stuttering dataset — disfluency-event classifier training (28,177 clips)
KSoF (Kassel State of Fluency)	arxiv.org/abs/2203.05383	Research	PhonNet: fine-grained disfluency-type classification, 6 categories (5,597 clips, 37 speakers)
LibriStutter	github.com/Kourkounakis/LibriStutter	Open	PhonNet: synthetic data augmentation for low-resource stuttering training
TORGO (Dysarthria)	cs.toronto.edu/~complingweb/data/TORGO	Free research	PhonNet: speech-production-disorder baseline (negative-class reference)
AI4Bharat IndicSUPERB	github.com/AI4Bharat/indic-superb	CC BY 4.0	Critical: Hindi/Marathi/Telugu speech and ASR base for all Indian-language phonological tasks
ChildVox Benchmark	arxiv.org/abs/2605.29257	Research	PhonNet: children's speech baseline across 17 child-centered audio datasets, 20+ sub-tasks
CHILDES	childes.talkbank.org	CC BY-NC-SA 3.0	PhonNet: child speech/language development norms, multiple languages incl. limited Hindi

6.6 Anxiety, Depression & Socio-Emotional Datasets

Dataset	Link	License	SEREN Use
---------	------	---------	-----------

DAIC-WOZ / E-DAIC	dcapswoz.ict.usc.edu (free academic registration)	Research (free)	EmotNet: depression/anxiety acoustic and behavioural feature calibration against PHQ-8 scores (189–275 interviews)
MODMA	modma.lzu.edu.cn/data/index	Free research	EmotNet: depression signal features (EEG + audio), 53 subjects
Mental Health Text (Hugging Face)	huggingface.co/datasets/ourafla/Mental-Health_Text-Classification_Dataset	CC BY (open)	EmotNet: language-pattern features for teacher-questionnaire NLP (48,945 labeled samples)
Sentiment Analysis MH (Kaggle)	kaggle.com/datasets/suchintikasarkar/sentiment-analysis-for-mental-health	Open (Kaggle)	EmotNet: expanded multi-label emotional-state classifier training
SCARED Normative Data	via pmc.ncbi.nlm.nih.gov (Birmaher et al., 1999)	Open in published papers	EmotNet: normative reference for anxiety-screening proxy scoring (ages 8–18)
SMQ Selective Mutism Norms	pmc.ncbi.nlm.nih.gov/articles/PMC7528533	CC BY	EmotNet: selective-mutism teacher-proxy scoring calibration

6.7 Multimodal & General Child-Development Datasets

Dataset	Link	License	SEREN Use
ABIDE I & II	fcon_1000.projects.nitrc.org/indi/abide	Open Access	Research reference for the ASD module (Phase 3 expansion), 1,112+ participants
AI4Bharat IndicNLP Corpus	github.com/ai4bharat/indic-nlp-library	MIT License	PhonNet: Indian-language tokenization and phoneme-mapping base, 12 languages
IIT Bombay Hindi Corpus	cfilt.iitb.ac.in/iitb_parallel	CC BY-NC	PhonNet: Hindi language-model base for phonological task generation (1.49M sentence pairs)

What does not yet exist — stated plainly: for most of the Insecurities & Confidence Deficits category (21 conditions) and much of the Silent Profiles category (11 conditions), no dedicated public sensor/behavioural dataset exists anywhere in the world. SEREN's approach is to build detection tasks around already-validated psychological scales (e.g., the Clance Impostor Phenomenon Scale, DERS for Emotional Dysregulation), use the general-purpose models above as feature extractors, and fine-tune a small final classification layer on SEREN's own consented pilot data. Treat the accuracy figures for this category as provisional until the pilot study validates them.

7. Full Condition → Dataset → Detection-Modality Map

Complete mapping from each of SEREN's 84 condition/subtype targets to its SEREN AI module, primary training datasets, and the specific detection modality used.

#	Condition	Module	Primary Datasets	Detection Modality
1	Dyslexia (Phonological)	GazeNet+PhonNet	ETDD70, FluencyBank	Eye-tracking reading regressions + phonological task RT
2	Dyslexia (Surface)	DrawNet+PhonNet	Dyslexia HW (Kaggle), UCLASS	Handwriting CNN + reading speed
3	Dyslexia (Deep/Mixed)	All 3 modules	ETDD70 + HW datasets + FluencyBank	Full multimodal fusion
4	Dyscalculia (Core)	NumNet	NSS Norms, PISA Math	Number comparison RT + subitizing accuracy
5	Dyscalculia (Access deficit)	NumNet+SpatialNet	PISA Math subsets	Symbolic processing + spatial tasks
6	Dysgraphia (Phonological)	DrawNet+PhonNet	IAM, Devanagari (Kaggle)	Handwriting kinematics + spelling audio
7	Dysgraphia (Motor)	DrawNet	IAM, CASIA, ICDAR	Stroke velocity, pressure, jerk CNN
8	Mixed SLD (Dyslexia+Dyscalculia)	FusionNet	All LD datasets combined	Full multimodal ensemble
9	Mixed SLD (Dyslexia+Dysgraphia)	FusionNet	ETDD70 + HW datasets	Handwriting + eye-tracking fusion
10	ADHD — Inattentive	AttentNet+GazeNet	IEEE EEG ADHD, ADHD-200	Sustained attention task + gaze deviation
11	ADHD — Hyperactive-Impulsive	AttentNet	IEEE EEG ADHD, KSoF	Response inhibition + commission errors
12	ADHD — Combined	AttentNet+FusionNet	IEEE EEG + ADHD-200 + Mendeley EEG	Full attention battery
13	ADHD in Adults	AttentNet	Mendeley EEG ADHD Adults	Adult cognitive task battery
14	ADHD + Dyslexia Comorbid	FusionNet	ADHD-200 + ETDD70	Joint feature fusion
15	Processing Speed Disorder	FusionNet (latency)	Age-normed latency benchmarks	Response latency percentile (all tasks)
16	Working Memory Deficit	AttentNet	ADHD-200, EEG cognitive sets	Sequence recall + digit-span proxy
17	Dev. Coordination Disorder	DrawNet	MABC-2 norms, IAM	Motor stroke kinematics (jerk)
18	Dyspraxia (Planning)	DrawNet	MABC-2, ICDAR	Sequential motor task timing

19	Visual-Motor Integration	DrawNet+SpatialNet	Raven's Progressive Matrices norms	Shape reproduction + spatial tasks
20	Stuttering (Repetitions)	PhonNet	UCLASS, FluencyBank, SEP-28k	Audio: repetition event detection
21	Stuttering (Prolongations)	PhonNet	UCLASS, KSoF, LibriStutter	Audio: prolongation acoustic feature
22	Stuttering (Interjections)	PhonNet	FluencyBank, LibriStutter	Audio: interjection token detection
23	Cluttering (Fast Speech)	PhonNet	FluencyBank (rate analysis)	Audio: syllable rate + intelligibility
24	Dysarthria	PhonNet	TORGO, ChildVox	Audio: articulation + prosody
25	Childhood Apraxia of Speech	PhonNet	ChildVox, CHILDES	Audio: inconsistent error pattern
26	Expressive Language Disorder	PhonNet	CHILDES, ChildVox	Audio: MLU + vocabulary diversity
27	Receptive Language Disorder	PhonNet+AttentNet	CHILDES, ChildVox	Response to instruction accuracy + latency
28	Word-Finding Difficulty	PhonNet	FluencyBank + fluency norms	Verbal fluency: items/60s + pause rate
29	Phonological Disorder	PhonNet	ChildVox, CHILDES	Phoneme error rate + consistency
30	Auditory Processing Disorder	PhonNet	ChildVox (noise-added), SCAN-3 norms	Target-in-noise accuracy degradation
31	Voice Disorders	PhonNet	TORGO, ChildVox	Acoustic voice-quality features
32	Generalised Anxiety (GAD)	EmotNet	DAIC-WOZ, SCARED norms	Task abandonment + latency + teacher proxy
33	Social Anxiety	EmotNet	SCARED norms, SMQ	Teacher proxy: oral participation rate
34	Separation Anxiety	EmotNet	SCARED norms, teacher proxy	Teacher proxy: school-avoidance markers
35	Test / Performance Anxiety	EmotNet+AttentNet	DAIC-WOZ, SCARED	Timed vs. untimed accuracy delta
36	Selective Mutism	EmotNet	SMQ (Oerbeck 2020), SCARED	Teacher proxy: verbal-utterance frequency
37	Specific Phobia (School)	EmotNet	SCARED norms	Teacher proxy: avoidance behaviour
38	Depression Indicators	EmotNet	DAIC-WOZ, MODMA, Reddit MH	Task engagement withdrawal markers
39	Social Withdrawal	EmotNet	Teacher proxy + task performance	Performance-participation discrepancy flagging
40	Introversion-Driven Suppression	EmotNet+SpatialNet	Pilot data + teacher proxy	Hidden strength + participation gap
41	Home Environment Stress	EmotNet	Teacher proxy (structured questions)	Structured stress indicators
42	Bullying Victimization	EmotNet	Teacher proxy + avoidance signals	Task-abandonment spike + withdrawal

43	Trauma Response	EmotNet	DAIC-WOZ trauma subset	Hypervigilance proxy: startle response
44	ASD — Social (Phase 3)	EmotNet	ABIDE, ChildVox (Phase 3)	Facial AU analysis + social task engagement
45	ASD — Communication (Ph.3)	PhonNet	ChildVox, CHILDES (Phase 3)	Pragmatic language feature analysis
46	ASD — Sensory (Ph.3)	AttentNet	ABIDE + ChildVox (Phase 3)	Sensory task response pattern analysis
47	Twice Exceptional (2e)	SpatialNet+FusionNet	Raven's norms + SLD datasets	High spatial + low reading discrepancy
48	Hidden Spatial Intelligence	SpatialNet	Raven's CPM norms	Matrix completion speed + accuracy above norm
49	Hidden Pattern Recognition	SpatialNet	Raven's + custom patterns	Sequence completion RT below norm
50	Kinesthetic Intelligence	DrawNet+teacher proxy	MABC-2 + teacher proxy	High motor accuracy despite low academics
51	Verbal IQ Suppressed by Language Barrier	PhonNet+EmotNet	CHILDES + teacher proxy	Comprehension vs. oral expression gap
52	Mathematical Anxiety	NumNet+EmotNet	PISA Math, SCARED math subscale	Performance collapse under timed math
53	Arithmetic Fact Retrieval (Isolated)	NumNet	NSS norms	Fact RT without magnitude deficit
54	Place Value Confusion	NumNet	Teacher proxy + custom tasks	Error pattern on place-value items
55	Fraction/Ratio Deficit (Age 10+)	NumNet	PISA Math subsets	Custom ratio task vs. norms
56	Reading Comprehension (No Decoding Deficit)	GazeNet+PhonNet	ETDD70 + ChildVox	Gaze patterns on comprehension questions
57	RAN Deficit	PhonNet	RAN norms + FluencyBank	Object/letter naming speed vs. age norm
58	Orthographic Processing	DrawNet+GazeNet	IAM + ETDD70	Sight-word latency + letter CNN
59	Reading Fluency (No Accuracy Deficit)	GazeNet+PhonNet	ETDD70 + ChildVox	WPM vs. age norm; regression rate
60	Spelling Disorder	DrawNet+PhonNet	IAM + phonological norms	Phoneme-grapheme mapping error analysis
61	Executive Function: Planning	AttentNet	ADHD-200 + EEG cognitive sets	Multi-step sequencing accuracy
62	Response Inhibition Deficit	AttentNet	IEEE EEG ADHD + Go/No-Go norms	Commission error rate
63	Cognitive Flexibility Deficit	AttentNet	ADHD-200 + Trail Making norms	Task-switching accuracy drop

64	Verbal Working Memory	AttentNet+PhonNet	Digit span norms	Digit span + sentence repetition
65	Visual-Spatial Working Memory	SpatialNet+AttentNet	Corsi block norms + Raven's	Spatial sequence reproduction span
66	Auditory Discrimination	PhonNet	ChildVox + SCAN-3 norms	Phoneme pair discrimination RT
67	Visual Discrimination	DrawNet+GazeNet	Shape discrimination norms	Shape matching accuracy + fixation
68	Tactile Processing (proxy)	DrawNet	Motor task pressure + DCD norms	Touchscreen pressure variability
69	Pragmatic Language Disorder (Ph.3)	PhonNet+EmotNet	ChildVox, CHILDES (Phase 3)	Turn-taking proxy + topic maintenance
70	Theory of Mind Deficit (Ph.3)	EmotNet	ChildVox + social tasks (Phase 3)	Social scenario response analysis
71	Non-Verbal Learning Disability	SpatialNet+DrawNet	Raven's + MABC-2	Verbal vs. visual-spatial discrepancy
72	Emotional Dysregulation	EmotNet+AttentNet	DAIC-WOZ + Reddit MH	Task frustration abandonment spikes
73	Low Frustration Tolerance	EmotNet+AttentNet	Teacher proxy + error patterns	Error-triggered abandonment rate
74	Impulsivity (Non-ADHD)	AttentNet	Go/No-Go + teacher proxy	Commission errors without attention deficit
75	Perfectionism (Maladaptive)	EmotNet	DAIC-WOZ + Reddit Anxiety	Self-correction rate + latency on easy items
76	Fine Motor Delay	DrawNet	MABC-2 norms	Stroke smoothness + speed vs. age
77	Gross Motor Delay (proxy)	EmotNet (teacher)	Teacher proxy only	Teacher-reported motor observation
78	Handedness Confusion	DrawNet	Custom stroke direction data	Stroke hand-dominance inconsistency
79	Adult Dyslexia	GazeNet+PhonNet	FluencyBank adults + PMC 2024 study	Slow reading speed + reduced regression
80	Adult Dyscalculia	NumNet	NSS adult norms + PISA adult	Arithmetic RT vs. adult norms
81	Adult ADHD	AttentNet	Mendeley EEG ADHD Adults	Sustained attention + working-memory battery
82	Adult Word-Finding	PhonNet	FluencyBank + fluency norms	Category fluency score + pause rate
83	Reading Comprehension (Adult)	GazeNet	ETDD70 + FluencyBank adult	Gaze regression + reading speed vs. adult norm
84	Processing Speed (Adult)	FusionNet	All tasks, adult latency norms	Response latency percentile vs. adult norms

8. Training Protocols — Environment, Download Commands & Model Code

Executable protocols for SEREN's AI development team, Python 3.10+ compatible.

8.1 Environment Setup

```
pip install tensorflow torch torchaudio transformers scikit-learn
pip install mediapipe opencv-python pandas numpy matplotlib
pip install kaggle zenodo-get datasets huggingface-hub
```

8.2 Downloading Key Datasets

```
# ETDD70 (Eye-Tracking, Dyslexia) — Zenodo
pip install zenodo-get
zenodo_get 13332134 # ~2GB: CSV gaze coordinates, fixation events, saccade data

# Dyslexia Handwriting Dataset — Kaggle
kaggle datasets download -d drizasazanitaisa/dyslexia-handwriting-dataset
# 78k normal + 52k reversal + 8k corrected letter images

# IEEE EEG ADHD Children — Kaggle mirror
kaggle datasets download -d danizo/eeg-dataset-for-adhd
# 19-channel EEG CSV, 61 ADHD + 60 control children, ages 7-12

# UCLASS Stuttering — register free at uclass.org.uk
# 139 WAV monologues, 81 people who stutter, ages 5-47, CHAT transcripts

# FluencyBank — fluency.talkbank.org
pip install pylangacq # for reading CHAT transcript format

# AI4Bharat IndicSUPERB — Hindi + Marathi speech
from datasets import load_dataset
ds_hi = load_dataset('ai4bharat/indicsuperb', 'hi')
ds_mr = load_dataset('ai4bharat/indicsuperb', 'mr')

# DAIC-WOZ — register free academic access at dcapswoz.ict.usc.edu
# ~50GB: audio interviews + PHQ-8 scores + transcripts
```

8.3 DrawNet — Handwriting/Drawing Protocol

```
# Step 1: Load Kaggle dyslexia HW images, resize 224x224 for EfficientNetB0
# Labels: 0=normal, 1=reversal, 2=corrected
# Augment: rotation +-10deg, brightness +-0.2 (no horizontal flip)

base = tf.keras.applications.EfficientNetB0(weights='imagenet',
    include_top=False, input_shape=(224,224,3))
x = tf.keras.layers.GlobalAveragePooling2D()(base.output)
x = tf.keras.layers.Dense(128, activation='relu')(x)
out = tf.keras.layers.Dense(3, activation='softmax')(x)
model = tf.keras.Model(base.input, out)
model.compile(optimizer='adam', loss='sparse_categorical_crossentropy')

# Stroke kinematics from touchscreen MotionEvent data:
# velocity[i] = sqrt((dx/dt)^2 + (dy/dt)^2)
# jerk[i] = abs(velocity[i]-velocity[i-1]) / dt
# pressure = touchscreen contact area
# Normalize per age band (6-8, 9-11, 12-14) using pilot-school norms
```



```
# CRITICAL – Hindi adaptation:
# Pre-train on kaggle.com/datasets/suvo00/hindi-character-recognition
# Fine-tune CNN on Hindi letters BEFORE dyslexia transfer learning
```

8.4 GazeNet — Eye-Tracking Protocol

```
# MediaPipe Iris landmarks (indices 468-477) for gaze direction
# Feature extraction from ETDD70:
df = pd.read_csv('ETDD70/participant_01_gaze.csv')
features = {
    'mean_fix_dur': df[df.type=='fixation']['duration'].mean(),
    'regression_n': len(df[(df.type=='saccade') & (df.x_end < df.x_start)]),
    'saccade_amp': df[df.type=='saccade']['amplitude'].mean(),
    'fixation_count': len(df[df.type=='fixation']),
}

model = tf.keras.Sequential([
    LSTM(128, return_sequences=True, input_shape=(timesteps, 6)),
    LSTM(64), Dropout(0.3),
    Dense(32, activation='relu'),
    Dense(1, activation='sigmoid') # dyslexia risk probability
])
```

8.5 PhonNet — Speech/Phonological Protocol

```
from transformers import Wav2Vec2ForCTC, Wav2Vec2Processor
model = Wav2Vec2ForCTC.from_pretrained('ai4bharat/indicwav2vec-hindi')
processor = Wav2Vec2Processor.from_pretrained('ai4bharat/indicwav2vec-hindi')

# response_latency = time(first_speech_onset) - time(stimulus_offset)
# phoneme_accuracy = transcription match rate to expected phoneme
# fluency_rate = phonemes_per_second (RAN task scoring)

# Stuttering fine-tune on UCLASS WAV + CHAT transcripts:
import pylangacq
data = pylangacq.read_chat('uclass/monologue_01.cha')
# Labels per frame: REP/PROL/INT/FLUENT
# Fine-tune 80% UCLASS train / 20% validate; export as TFLite
```

8.6 AttentNet — ADHD/Attention Protocol

```
# Behavioural proxies only – no EEG hardware required on-device:
sustained_attn_miss = missed_targets / total_targets
commission_error = false_alarms / no_go_trials
rt_variability = df['rt'].std() / df['rt'].mean()
gaze_off_task_pct = off_task_frames / total_frames # via GazeNet

# Normative calibration (age-banded z-scores):
z_score = (child_score - age_norm_mean) / age_norm_sd
# Flag: z_score > 2.0 on attention battery = ADHD risk flag
```

8.7 EmotNet — Anxiety/Emotional Signals Protocol

```
from transformers import pipeline
clf = pipeline('text-classification', model='google/muril-base-cased')
# Teacher free-text (Hindi/Marathi) -> anxiety_flag / withdrawal_flag / home_stress_flag

def emotion_flags(task_data, baseline):
    flags = []
    if task_data['abandon_rate'] > 0.25: flags.append('ANXIETY_AVOIDANCE')
    if task_data['self_correct_rate'] > 3*baseline['self_correct']:
```

```
flags.append('PERFECTIONISM_ANXIETY')
    if task_data['easy_item_latency'] > 2*baseline['easy_latency']: flags.append('OVERTHINKING_ANXIETY')
    if (task_data['untimed_acc'] - task_data['timed_acc']) > 0.3: flags.append('PERFORMANCE_ANXIETY')
    return flags
```

8.8 On-Device Deployment (TFLite)

```
converter = tf.lite.TFLiteConverter.from_keras_model(trained_model)
converter.optimizations = [tf.lite.Optimize.DEFAULT]
converter.target_spec.supported_types = [tf.float16]
tflite_model = converter.convert()
open('seren_drawnet.tflite', 'wb').write(tflite_model)
# Target: <50MB per sub-model, <15ms inference on a mid-range Android device
# FusionNet runs on-device in Dart: <5ms, no internet required
```

All SEREN models run fully offline. No child data leaves the device during inference. Cloud sync is optional, consent-gated, and anonymised before transmission.

9. Accuracy Model, Validation Roadmap & Honest Limitations

9.1 How Fusion Drives Accuracy

Modalities Active	Expected Accuracy	Example	Fusion Method
1 modality only	65-75%	Dyslexia via eye tracking alone	Single-model output
2-3 modalities	78-85%	Dyslexia: eye tracking + handwriting + phonological	Weighted average, highest-confidence model leads
4-6 modalities	87-93%	ADHD: CPT + gaze + sensors + Go/No-Go	Random-forest ensemble, feature-importance weighted
7-8 modalities + longitudinal	93-99%	Full SEREN session with baseline comparison	Deep fusion model, on-device uncertainty quantification

9.2 Validation Roadmap

1. **Instrument calibration** — build age-stratified normative datasets from pilot deployment cohorts.
2. **Concurrent validity study** — compare SEREN outputs against formal clinician assessment on a matched sample.
3. **Longitudinal outcome study** — track intervention-arm vs. control-arm outcomes over one academic year.
4. **External peer-reviewed publication** of validation results.

State plainly to any investor or partner: the datasets in Section 6 provide real, verified pre-training signal for SEREN's detection modules — they are not a substitute for SEREN's own clinical validation study. Most are adult, non-Indian, or neuroimaging-based, and are used to validate the general feature-extraction pipeline rather than as India-specific, child-specific ground truth. That ground truth comes only from SEREN's own consented pilot cohort.

10. Reference Bibliography

- Birmaher, B. et al. (1999). SCARED normative data. *JAACAP*, 38(10), 1230–1236.
- Bayerl, S.P. et al. (2022). KSoF: Kassel State of Fluency Dataset. arXiv:2203.05383.
- Clance, P.R., & Imes, S.A. (1978). The Imposter Phenomenon in High Achieving Women. *Psychotherapy: Theory, Research & Practice*, 15(3).
- Dostalova, N. et al. (2024). ETDD70: Eye-Tracking Dyslexia Dataset. Zenodo. doi.org/10.5281/zenodo.13332134
- Fayyad, J. et al. (2007). Cross-National Prevalence and Correlates of Adult ADHD. *British Journal of Psychiatry*, 190.
- Feng, T. et al. (2026). ChildVox: A Speech, Audio, and Large Audio-Language Model Benchmark. arXiv:2605.29257.
- Gratch, J. et al. (2014). DAIC-WOZ Database. dcapswoz.ict.usc.edu
- Joseph, A., & Devu, B.K. (2022). Prevalence and Pattern of Learning Disability in India. *Indian Journal of Psychiatric Nursing*, 19(2).
- Lea, C. et al. (2021). SEP-28k: A Dataset for Stuttering Event Detection from Podcasts. *ICASSP 2021*.
- MacWhinney, B. (2000). The CHILDES Project. childes.talkbank.org
- Ministry of Education, Government of India (2020). *National Education Policy 2020*.
- Ministry of Education, Government of India (2021). *NIPUN Bharat Mission*.
- Oerbeck, B. et al. (2020). Selective Mutism Questionnaire. PMC7528533.
- Ratner, N.B., & MacWhinney, B. (2018). FluencyBank. fluency.talkbank.org
- Rello, L., & Bayarri, C. (2020). Predicting Risk of Dyslexia with a Gamified Test. *PLOS ONE*.
- Sadeghi Bajestani, G. et al. (2023). EEG Signals from Adults with ADHD. Mendeley Data. doi.org/10.17632/6k4g25fhzg.1
- Sedmidubsky, J. et al. (2024). ETDD70: Eye-Tracking Dataset for Classification of Dyslexia. *SISAP 2024*.
- ADHD-200 Consortium (2012). fcon_1000.projects.nitrc.org/indi/adhd200
- ABIDE Consortium (2012). fcon_1000.projects.nitrc.org/indi/abide
- AI4Bharat (2022). IndicSUPERB & IndicNLP. github.com/AI4Bharat

SEREN — Every child is a star. Every adult deserves a second chance.

Founder: Sansakardeep Talikote | Confidential — Complete Single-Volume Research & Dataset Blueprint.